1 Team 4

Table 1: Members

Name	Email
Jackson Weibel(Team Leader)	jacksonweibel@tamu.edu
Miles Turnage	milesturnage@tamu.edu
Gabriel Rangel-Orozco	gabrielrangelorozco@tamu.edu
Gwilym Couch	gwilymcouch@tamu.edu
Vihaan Ketavarapu	vihaanket@tamu.edu

GitHub Repository: `https://github.com/JacksonWeibelDev/DAEN328-Term-Project`

2 Data Source

Our group selected the National Crime Victimization Survey (NVCS) Personal Victimization dataset, published by the Bureau of Justice Statistics.

- API: `https://api.ojp.gov/bjsdataset/v1/gcuy-rt5g.json`
- This is our own proposal.
- No api key is needed.
- The survey is conducted annually, with historical data from 1993 to 2024.
- The API supports pagination using limit and offset parameters.
- Each entry has a unique identifier(idper) we will use for incremental pulls.

3 Proof it works

```
weibel@JacksonsLaptop:/mnt/c/Dev/DAEN328/DAEN328-Term-Project$ python3 extract.py
{'Status': 200, 'Records': 68852, 'Columns': ['idper', 'yearq', 'year', 'ager', 'sex', 'hispanic', 'race
', 'race_ethnicity', 'hincome1', 'hincome2', 'marital', 'popsize', 'region', 'msa', 'locality', 'educatn
1', 'educatn2', 'veteran', 'citizen', 'newcrime', 'newoff', 'seriousviolent', 'notify', 'vicservices', '
locationr', 'direl', 'weapon', 'weapcat', 'injury', 'serious', 'treatment', 'offenderage', 'offendersex'
, 'offtracenew', 'wgtviccy', 'series', 'newwgt']}
```

Figure 1: Successful NCVS API request returning 68,852 records and 37 columns.

4 Data description

Our project will analyze crime victimization patterns in the United States, focusing on the relationship between crime characteristics and whether incidents are reported to police. The goal is to identify patterns that could help researchers better understand unreported victimization.

Our dashboard will answer five questions:

1. Which types of victimizations are most and least likely to be reported?
2. How have police reporting patterns changed over time?
3. How does the victim-offender relationship relate to police reporting?
4. Are victimizations involving weapons or injuries more frequently reported?
5. How do reporting patterns differ by age group and region?

Our proposed PostgreSQL tables:

- **Respondents:** Stores demographic and background information about surveyed individuals.

  Columns: `idper`, `yearq`, `year`, `ager`, `sex`, `hispanic`, `race`, `race_ethnicity`, `hincome1`, `hincome2`, `marital`, `popsize`, `region`, `msa`, `locality`, `educatn1`, `educatn2`, `veteran`, `citizen`.

  Proposed composite primary key: `(idper, yearq)`.

- **Offense Types:** Stores crime classifications and descriptions.

  Columns: `newoff`, `newcrime`, `seriousviolent`, and `offense_name` (derived from the codebook).

  Primary key: `newoff`.

- **Victimizations:** Stores individual victimization records, including reporting status, crime circumstances, offender characteristics, and survey weights.

  Columns: `victimization_id` (source `:id`), `idper`, `yearq`, `newoff`, `notify`, `vicservices`, `locationr`, `direl`, `weapon`, `weapcat`, `injury`, `serious`, `treatment`, `offenderage`, `offendersex`, `offtracenew`, `wgtviccy`, `series`, `newwgt`, `updated_at` (source `:updated_at`).

  Provisional primary key: `victimization_id`.

  Foreign keys: `(idper, yearq)` references Respondents; `newoff` references Offense Types.

5 Plan and Roles

Table 2: Members

Name	Role
Jackson Weibel	Docker, Loading
Miles Turnage	Extraction
Gabriel Rangel-Orozco	Dashboard
Gwilym Couch	Shadowing
Vihaan Ketavarapu	Transformation

Project Risk: One potential challenge is the dataset's incremental update behavior. Our tests showed that records from multiple survey years shared the same update timestamp, suggesting that BJS may refresh data in bulk.

To address this, we plan to track the last successful update timestamp, use PostgreSQL UPSERT operations to prevent duplicate records, and validate record identifier stability.